

Improving Robotic Imitation Learning with Predicted Facial Motion Using Transformers

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Abstract— This study proposes a Transformer-based approach with cross-attention for predicting human facial movements in face-related robotic control tasks and integrating these predictions into an imitation learning framework. A dataset of human facial videos was constructed, and landmarks were extracted using the MediaPipe framework. Three prediction methods were compared, and the cross-attention model achieved the best performance in both landmark localization accuracy and image quality. In imitation learning experiments, facial motion trajectories sampled from real human data trajectories were used, and the success rate increased from 42% to 60% and ultimately to 74% when predicted landmarks were incorporated.

Keywords— Transformer, Cross-attention, Facial Motion Prediction, Imitation Learning, Robotic Feeding

I. INTRODUCTION

Recent advancements in Human-Robot Interaction (HRI) have enabled robotic arms to be increasingly deployed in healthcare and assistive applications for the elderly and disabled [1]. Assistive robots require precise face-related operations such as feeding. [2]. Such interactions demand both high task accuracy and strict safety guarantees [3].

However, conventional robotic motion control still depends on static facial landmark information, which limits adaptability in dynamic situations. Since human facial movements can change subtly during interaction, this often causes planning errors or even safety risks [4] [5].

To overcome these limitations, this study employs a cross-attention Transformer to predict short-term facial motion and integrate these predictions into the Action Chunking with Transformers (ACT) imitation learning framework [6].

The main contribution of this study is the integration of cross-attention Transformer predictions into ACT-based imitation learning.

II. RELATED WORK

Imitation learning (IL) has proven effective for robotic control in high-dimensional action spaces [7]. Common methods such as Behavior Cloning [8] and Inverse Reinforcement Learning [9] have been applied to trajectory generation and manipulation tasks. However, these approaches rely on static input states, limiting their performance in dynamic human-robot interaction scenarios.

Transformers, originally developed for natural language processing, have shown strong capability in modeling long-term dependencies and have been applied to motion prediction, trajectory generation, and robot planning [10]–[15]. Yet, their

use for facial dynamics prediction in assistive tasks remains underexplored.

Facial landmark detection and tracking are essential for face-related tasks such as feeding and wiping [16]. While CNN-based and RNN-based models can detect and model temporal continuity, they often fail under occlusion or rapid head motion [17][18].

The Action Chunking with Transformers (ACT) framework [19] effectively learns temporal action patterns and generates long-horizon motion sequences. Combining ACT with Transformer-based facial motion prediction is expected to improve adaptive robot control in dynamic face-related scenarios [20].

In summary, while prior studies have advanced imitation learning, Transformer-based sequence modeling, and facial landmark prediction, there is still a gap in integrating Transformer-predicted facial motion into imitation learning frameworks for adaptive control in face-related robotic tasks. This study aims to address this gap by integrating Transformer-predicted facial motion into an imitation learning framework.

III. APPROACH

The proposed system consists of two main modules: (1) a Transformer-based facial motion prediction model, and (2) an imitation learning-based control model (ACT).

As shown in Fig. 1, RGB images from three cameras and the robot's joint states are fed into the ACT control policy. The Transformer module uses the front-facing camera to predict future facial landmarks for several frames ahead. These predictions are incorporated into the ACT policy as observation inputs, allowing the robot to plan adaptive face-related motions.

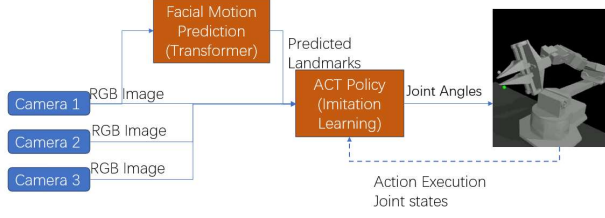


Figure 1. Overall architecture of the proposed system. Facial motion prediction provides future facial landmarks to the ACT policy, which outputs the robot trajectory.

A. Facial Motion Prediction Using Transformer

Transformers effectively model long-range dependencies and are applied here to predict sequential facial landmark motions. In this study, we leverage the sequential modeling capability of Transformers to predict the temporal evolution of facial landmarks over several future frames, making them particularly suitable for capturing dynamic variations in facial landmark trajectories.

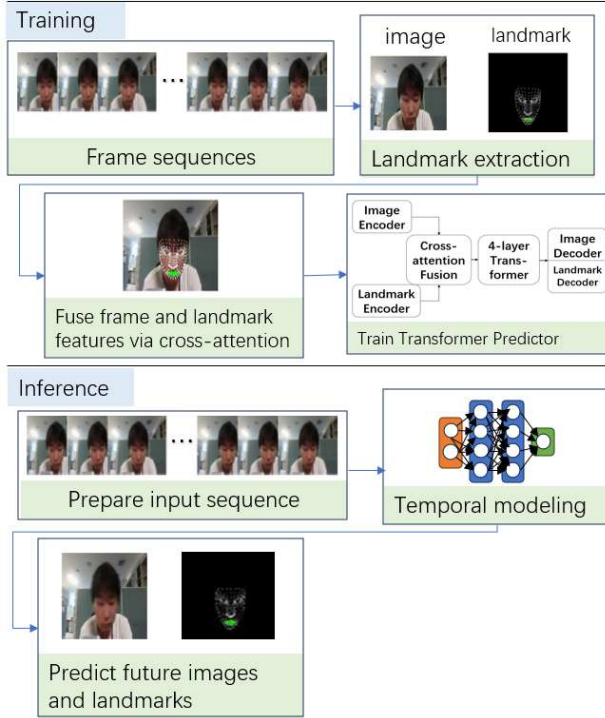


Figure 2. Training and inference pipeline of the facial motion prediction module.

The input is a sequence of RGB frames and normalized 2D facial landmarks; the mouth region is particularly important for predicting subtle facial motions. In this work, we predict normalized 2D landmark coordinates (U, V) , where u and v are image-space positions used for normalization.

In addition to predicting mouth landmarks, the model also predicts the corresponding RGB frames as an auxiliary

supervision signal. This encourages the Transformer to learn more robust temporal features, even though only landmark predictions are ultimately used for robotic control. Moreover, predicting the full set of landmarks, rather than just the mouth center, provides richer geometric information and results in more stable motion features. The mouth center used for control can then be derived from these predicted landmarks.

A cross-attention mechanism fuses visual and landmark embeddings before feeding them into the Transformer. The overall training and inference pipeline of the facial motion prediction module is illustrated in Fig. 2.

Mathematically, let the feature matrix of the image sequence be denoted as $I \in R^{T \times D_i}$, where T is the number of frames and D_i is the feature dimension of the image embeddings. Similarly, let the landmark feature matrix be $L \in R^{T \times D_l}$, where D_l is the feature dimension of the landmark embeddings. The cross-attention mechanism computes the attention weights between the landmarks and the image features as follows:

$$A = \text{softmax} \left(\frac{L \cdot I^T}{\sqrt{d}} \right) \quad (1)$$

where d is the dimension of the projected feature vectors used in the attention computation. These attention weights are then used to compute a combined context representation:

$$C = A \cdot I \quad (2)$$

Specifically, for each time step i , the fused feature vector c_i is given by:

$$c_i = \sum_{j=1}^T \text{softmax} \left(\frac{L_i I_j^T}{\sqrt{d}} \right) \cdot I_j \quad (3)$$

where L_i and I_j are the landmark and image feature vectors at time steps i and j , respectively. Note that L and I refer to the complete sequences.

The fused representations are input to the Transformer, and the model is trained with MSE loss as follows:

$$\mathcal{L}_{MSE} = \frac{1}{T} \sum_{t=1}^T \|\hat{y}_t - y_t\|^2 \quad (4)$$

where $\hat{y}_t$ and y_t denote the predicted and ground truth landmark coordinates at time step t , respectively.

B. Imitation Learning Framework

At the control level, this study adopts the Action Chunking with Transformers (ACT) framework, which maps system states to sequences of short action chunks using Transformer models.

In face-related robotic tasks, the goal is to safely and accurately interact with human faces. However, direct human experiments pose ethical and safety concerns, and collecting real-world demonstrations is time-consuming.

To address these issues, experiments are conducted in simulation, which ensures safety, enables fast iteration, and facilitates early validation of new methods before real-world transfer.

We integrate Transformer-predicted facial motion into simulation-based imitation learning. The Transformer trained on real facial videos predicts future mouth dynamics, which are incorporated into ACT for observation construction, policy inference, and trajectory generation (Fig. 3).

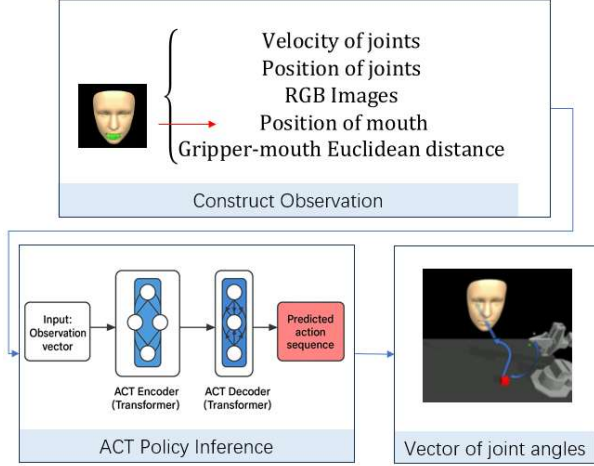


Figure 3. ACT-based imitation learning workflow from demonstration recording to trajectory output.

Since the simulated environment lacks realistic motion, prediction error statistics from real data are used to perturb simulated observations, improving realism.

This design allows the robot to adapt to facial motion uncertainty during imitation learning.

IV. EXPERIMENTS AND RESULTS

This section presents the experiments and designs performed in this study, covering Transformer-based prediction and the ACT imitation learning experiments.

A. Facial Motion Prediction Using Transformer

Facial motion videos were recorded at 30 FPS, covering various head movements and facial expressions. Each sequence contained 75 frames with RGB images and 2D landmarks extracted via MediaPipe, resized to 256×256 px for training and evaluation. The network combines a CNN encoder, a linear landmark encoder, and a cross-attention module followed by a 4-layer Transformer to predict future frames. Facial landmarks were extracted using the MediaPipe framework [21], generating the data for both training and evaluation. A Transformer-based network was implemented using PyTorch to jointly predict facial images and landmarks. RGB images were resized to 256×256 px, and 468 facial landmarks were extracted per frame via MediaPipe. The training objective combines image reconstruction and landmark regression losses as follows:

$$Loss_{total} = Loss_{img} + 0.5 \times Loss_{landmark} \quad (5)$$

We evaluated three approaches for Transformer-based facial motion prediction: 1. Image-only prediction (predicting frames then landmarks), 2. Simple fusion (jointly predicting from concatenated embeddings), 3. Cross-attention fusion (our proposed method integrating spatial and landmark features).

B. Transformer Prediction Results

Figures 4–6 visualize the predicted results of three approaches for comparison of image and landmark alignment.

Approach 1 produced blurry images, making facial landmarks difficult to extract (Fig. 4).

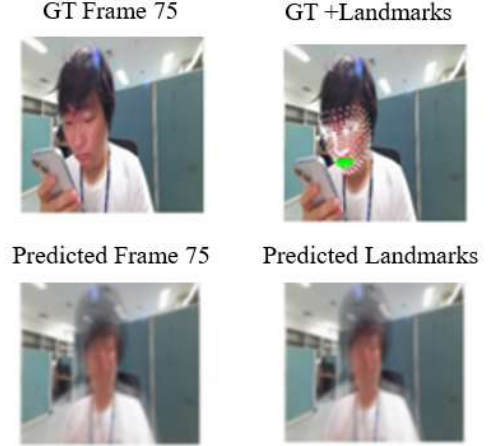


Figure 4. Visualization of prediction results for Approach 1, which directly predicts future images and then extracts facial landmarks from the predicted images.

Approach 2 slightly improved image clarity but still suffered from scattered landmark predictions (Fig. 5).

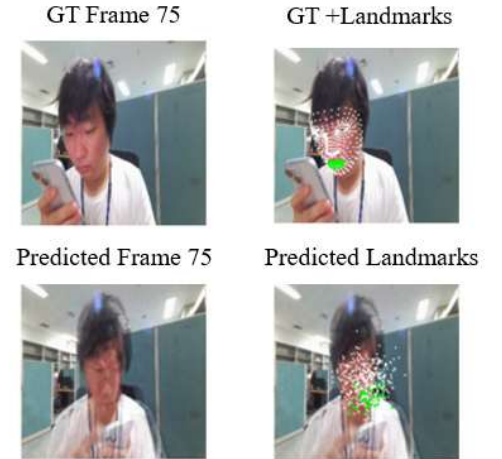


Figure 5. Visualization of prediction results for Approach 2, which employs a simple fusion strategy to simultaneously predict future images and facial landmarks from combined image and landmark features.

Approach 3 achieved the most accurate and consistent landmark localization through cross-attention fusion (Fig. 6).

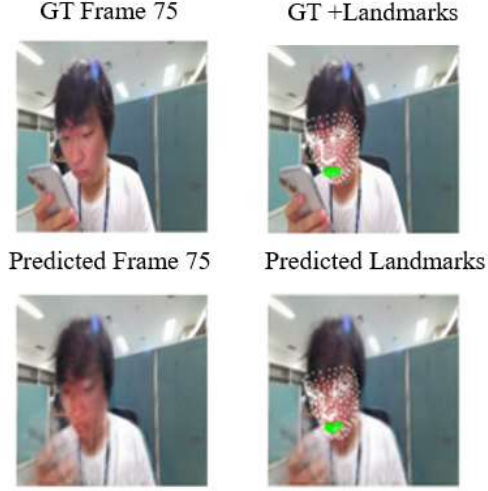


Figure 6. Visualization of prediction results for Approach 3, which introduces a cross-attention mechanism to integrate image features and landmark features for joint prediction.

TABLE I. COMPARISON OF PREDICTION PERFORMANCE

Metric	Approach		
	Approach 1	Approach 2	Approach 3
Mean ΔU	—	0.032	0.015
Var ΔU	—	0.061	0.013
Mean ΔV	—	0.036	0.017
Var ΔV	—	0.130	0.025
Average SSIM	0.662	0.818	0.928

Quantitative results in Tables I and II confirm that Approach 3 consistently outperformed the others in both mean/variance of mouth displacement (ΔU , ΔV) and SSIM.

TABLE II. PER-FRAME PREDICTION PERFORMANCE OF APPROACH 3

Metric	Predicted Frame				
	Frame 71	Frame 72	Frame 73	Frame 74	Frame 75
Mean ΔU	0.010	0.019	0.013	0.020	0.014
Var ΔU	0.010	0.014	0.014	0.013	0.014
Mean ΔV	0.013	0.021	0.015	0.017	0.017
Var ΔV	0.021	0.027	0.026	0.025	0.027
Average SSIM	0.926	0.927	0.928	0.928	0.929

C. ACT Imitation Learning under Varying Observation Strategies

The imitation learning experiments were conducted in the RoboManipBaselines[22] simulation environment, where the robot performs a feeding task guided by predicted facial motion. Figure 7 shows the setup used.

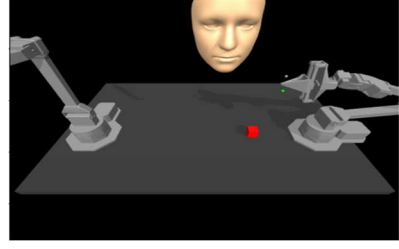


Figure 7. Visualization of the simulated imitation learning environment in RoboManipBaselines.

A scripted policy (Algorithm 1) was designed to generate demonstration trajectories ensuring consistent motion patterns for training.

Algorithm 1 SCRIPTED POLICY FOR FEEDING TASK

```

// Initialization
1: Initialize robot_arm at initial pose
2: Initialize time_step as dt = 0.02s
3: Initialize face_model at random initial position (base position +
   random offset within  $\pm 0.05$  m along X, Y, and Z)
// Define facial motion trajectory
4: Randomly select a start time t0 within the pre-recorded trajectory
   and extract a 6-second segment
5: target_position = initial_face_position + (trajectory(t0+6) -
   trajectory(t0))
// Phase 1: Approach and grasp cube
6: Move robot_arm toward cube position
7: Open gripper
8: Close gripper to grasp the cube
// Phase 2: Move toward mouth (face moves along face_velocity)
9: t  $\leftarrow$  0
10: while (|gripper.x - mouth_position.x|  $\geq$  0.03 m OR
        |gripper.y - mouth_position.y|  $\geq$  0.01 m OR
        |gripper.z - mouth_position.z|  $\geq$  0.01 m AND
        t  $\leq$  6s) do:
11:   move_velocity = (current_mouth_position -
        current_gripper_position) / (6 - t)
12:   Move face_model according to pre-sampled trajectory
13:   Move gripper by move_velocity * dt
14:   t  $\leftarrow$  t + dt
15: end while
// Phase 3: Insert cube into mouth and release
16: Open gripper to release the cube
17: Retract robot_arm from mouth to initial position (face fixed)

```

Each observation configuration was evaluated over 50 trials, and success rates were measured as the proportion of successful feeding motions.

Three observation configurations were tested: (1) Baseline (no landmarks), (2) Current (current mouth position only), (3) Current + Prediction (future mouth positions up to 5 frames).

Spatial errors were computed from prediction deviations using the following relations:

$$\max_{\delta x} = 2Z \tan(FOV_x/2) \Delta U \quad (6)$$

$$\max_{\delta y} = 2Z \tan(FOV_y/2) \Delta V \quad (7)$$

Figure 8 shows example images captured from the three RGB cameras (top, angled, and front-close views) used in the baseline observation setting.

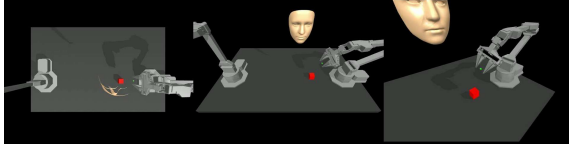


Figure 8. Example views from the three RGB cameras (top, angled, and front-close).

Experiments compared baseline, current, and prediction-based observations to evaluate whether predicted facial motion improves feeding success.

D. ACT Imitation Learning Results

Table III summarizes the success rates under different observation and prediction horizons. The baseline achieved 42%, while using current mouth position improved it to 60%. The best performance (74%) was obtained with a 2-frame prediction horizon, indicating that short-term facial motion prediction enhances policy adaptation.

TABLE III. SUCCESS RATES UNDER VARYING OBSERVATION STRATEGIES AND PREDICTION HORIZONS

Observation Type	Horizon (frames)	Success Rate	
Baseline	None	42%	
Current Only	0	60%	
Current + prediction		With Error	Without Error
	1	58%	66%
	2	72%	74%
	3	68%	72%
	4	62%	68%
	5	54%	60%

Table IV shows task completion times. Shorter prediction horizons (2–3 frames) led to faster and more stable executions, while longer horizons showed no further benefit.

TABLE IV. TASK COMPLETION TIMES UNDER DIFFERENT PREDICTION HORIZONS

Observation Type	Horizon (frames)	Completion Time	
Baseline	None	4.9	
Current Only	0	4.6	
Current + prediction		With Error	Without Error
	1	4.7	4.6
	2	4.3	4.3
	3	4.5	4.4
	4	4.5	4.4
	5	4.6	4.5

V. CONCLUSION

This study proposed a Transformer-based approach for predicting human facial movements and integrating these

predictions into an imitation learning framework for face-related robotic control tasks. The cross-attention Transformer achieved higher landmark prediction accuracy and better image quality than alternative methods.

In imitation learning experiments, incorporating facial motion information significantly improved task performance. Using facial trajectories extracted from real human demonstrations, the success rate improved from 42% to 60%, and further to 74% with predicted landmarks. The 2-frame prediction horizon produced the fastest task completion times under these realistic motion conditions.

These results demonstrate that integrating predicted facial motion into the observation space can substantially enhance imitation learning performance in dynamic human-robot interaction. Future work will focus on incorporating safety and collision modeling, enhancing goal-oriented imitation learning strategies, and extending evaluation from simulation to real-world scenarios to improve robustness and practical applicability.

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